

ICS 6B: Homework 3

1.12.5

(c)

$p = \text{get a job}$

$q = \text{buy a new car}$

$r = \text{buy a new house}$

The form is:

$\neg p \rightarrow \neg(q \wedge r)$

$\neg p$

—

$\therefore \neg q$

Hypothesis: $\neg p \rightarrow \neg(q \wedge r)$

Implication: $p \vee \neg(q \wedge r)$

De Morgan's Law: $p \vee \neg q \vee \neg r$

Hypothesis: $\neg p$

Thus: $F \vee \neg q \vee \neg r$

Identity Law: $\neg q \vee \neg r$

Thus its not valid because $\neg r$ could be true.

(d)

$p = \text{get a job}$

$q = \text{buy a new car}$

$r = \text{buy a new house}$

The form is:

$\neg p \rightarrow \neg(q \wedge r)$

$\neg p$

r

—

$\therefore \neg q$

Hypothesis: $\neg p \rightarrow \neg(q \wedge r)$

Identity Law: $p \vee \neg(q \wedge r)$

De Morgan's Law: $p \vee \neg q \vee \neg r$

Hypothesis: $\neg p$

Thus: $F \vee \neg q \vee \neg r$

Identity Law: $\neg q \vee \neg r$

Hypothesis: r

Thus: $\neg q \vee F$

Identity Law: $\neg q$
Thus it is valid.

1.13.1

(a)

$P(x)$ = x practices hard

$Q(x)$ = x plays badly

The form is:

$\forall x(P(x) \vee Q(x))$

$\exists x \neg P(x)$

—

$\therefore \exists x Q(x)$

Hypothesis: $\forall x(P(x) \vee Q(x))$

Let c be an arbitrary argument: $P(c) \vee Q(c)$

Hypothesis: $\exists x \neg P(x)$

Let d be an particular element: $\neg P(d)$

Since d is one of c , $P(c)$ is F : $F \vee Q(c)$

Implication Law: $Q(c)$

Universal: $\exists x Q(x)$

Thus it is valid.

1.13.4

(a)

Hypothesis: $\exists x(P(x) \wedge Q(x))$

Let c be an particular element: $P(c) \wedge Q(c)$

Existential Generalization: $\exists xP(x) \wedge \exists xQ(x)$

Thus valid.

(b)

Hypothesis: $\exists xP(x) \wedge \exists xQ(x)$

Let c be an particular element: Existential instantiation $P(c)$

Let d be an particular element: Existential instantiation $Q(d)$

Thus: $P(c) \wedge Q(d)$

Suppose

	P	Q
c	T	F
d	F	T

which satisfied $P(c) \wedge Q(d)$.

Thus, the conclusion $\exists x(P(x) \wedge Q(x))$ is False.

2.2.1

(a)

False, because there may be miss considered situation that makes the statement false, making it not universal.

2.2.5

(b)

Let $c = 3$

The sum of all the intergers smaller than c is $1 + 2 = 3$

Since $c = 3$, this is a legal statement.

(d)

$7c + 5d = 1$

Let $c = -2$ and $d = 3$

$7 * -2 + 5 * 3 = 1 = 1$

This is legal statement.

2.3.2

(a)

Variable reuse of k for $x = kw$ and $z = ky$.

(b)

> Let m be an integer such that $xz = m * wy$

This is wrong because introducing m from nowhere and assume that it is an integer is invalid.

(c)

Skipped step that $(kw)(jy) = (kj)(wy)$

And proving (kj) is an integer.

(d)

Not assuming k and w are integers, thus invalid.

2.4.3

(b)

Let x be a real number that $x \leq 3$. We will show that $12 - 7x + x^2 \geq 0$

Since $x \leq 3$, $x - 3 \leq 0$

Since $x \leq 3 < 4$, $x - 4 \leq 0$

Since two negative numbers multiplied together is a positive number, $(x - 3)(x - 4) \geq 0$

Simplify the left part: $12 - 7x + x^2 \geq 0$

■

2.4.4

(k)

It is true.

Assume we have x, y and z that $xy|z$, we will show that $x|z$ and $y|z$

Since $xy|z$, we assume there exists an integer k that $k * xy = z$ while $x \neq 0$ and $y \neq 0$

Thus, $(k * y) * x = z$

Since both k and y are integers, $(k * y)$ is an integer.

Therefore, since z equals x times other integer, and $z \neq 0$, $x|z$

Also, $(k * x) * y = z$

Since both k and y are integers, $(k * x)$ is an integer.

Therefore, since z equals x times other integer, and $z \neq 0$, $y|z$

■

(m)

It is false.

Let $x = 2$, $y = 3$ and $z = 5$.

Since $(y + z) = 8$, and $2 * 4 = 8$, so that $x|(y + z)$

However, there does not exist integer that allows $2 * k = 3$ or $2 * k = 5$

2.5.3

(c)

Assume we have real numbers x and y , y is not irrational and x is rational. We will show that xy is not irrational.

Since for a real number it can only be irrational or rational, and y is not irrational, y is rational.

Since x is a real number and y is a real number, xy is a real number.

Since y is a rational number and x is a rational number, xy is a rational number.

Therefore xy is not irrational.

■

2.6.6

(d)

Assume there exist a smallest integer x .

Case 1: $x > 0$

Therefore, $0 < x$. x is not smallest.

Case 2: $x = 0$

Therefore, any negative number is smaller than x . x is not smallest.

Case 3: $x < 0$

Since 2 and x are both integers, $2x$ is an integer.

Since $x < 0$, $2x < x$. x is not smallest.

There's a contradiction. Therefore, no smallest integer exists.

2.7.3

(d)

Case 1: x is positive and y is positive

$$x + y = x + y$$

which satisfied $|x + y| = x + y = |x| + |y|$

Case 2: x is negative and y is negative

$$-x - y = -(x + y)$$

which satisfied $|x + y| = -(x + y) = -x - y = |x| + |y|$

Case 3: x is positive and y is negative

Since $-y \leq |y|$ and $x \leq |x|$

$x + y \leq |x| + |y|$ when $x + y \geq 0$ or $|x + y| = -x - y \leq |x| + |y|$ when $x + y < 0$

Which both satisfied $|x + y| \leq |x| + |y|$

Case 4: x is negative and y is positive

Since $y \leq |y|$ and $-x \leq |x|$

$x + y \leq |x| + |y|$ when $x + y \geq 0$ or $|x + y| = -x - y \leq |x| + |y|$ when $x + y < 0$

Which both satisfied $|x + y| \leq |x| + |y|$

Thus proven.

